7.4 Matrices That Return to the Start

WORKSHEET

The question

STEP II 2020, Question 6

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Every 2×2 matrix satisfies $\mathbf{M}^2 - (\operatorname{tr}\mathbf{M})\mathbf{M} + (\det\mathbf{M})\mathbf{I} = \mathbf{0}$ (the Cayley–Hamilton theorem; check it by multiplying out). How could that help with questions about $\mathbf{M}^2$?

STAGE 1 The trace of the square

Covers: Part (i)

Ask yourself Multiply out $\mathbf{M}^2$ and add its diagonal entries. Compare with $(a+d)^2$.

STAGE 2 Square roots of $\pm\mathbf{I}$

Covers: Part (ii)

Ask yourself Substitute $\mathbf{M}^2 = \mathbf{I}$ into $\mathbf{M}^2 = t\mathbf{M} - d\mathbf{I}$, where $t = a + d$ and $d = \det \mathbf{M}$. If $t \neq 0$, what kind of matrix must $\mathbf{M}$ be?

Pause and try Show that $t \neq 0$ forces $\mathbf{M} = \pm\mathbf{I}$, which is excluded. So $t = 0$: what is $\det \mathbf{M}$? Do the same for $\mathbf{M}^2 = -\mathbf{I}$, and remember to prove the converses.

STAGE 3 Fourth roots

Covers: Part (iii)

Pause and try Put $\mathbf{N} = \mathbf{M}^2$, so $\mathbf{M}^4 = \mathbf{I}$ says $\mathbf{N}^2 = \mathbf{I}$. Use Part (ii) on $\mathbf{N}$. Why can $\det \mathbf{N}$ not be -1?

Ask yourself For $\mathbf{M}^4 = -\mathbf{I}$, use the second half of Part (ii) on $\mathbf{N}$, and write $\operatorname{tr}\mathbf{N}$ and $\det\mathbf{N}$ in terms of $\operatorname{tr}\mathbf{M}$ and $\det\mathbf{M}$ using Part (i).

STAGE 4 An eighth root of $\mathbf{I}$ that is not a rotation

Covers: Part (iv)

Ask yourself Any matrix with $\mathbf{M}^4 = -\mathbf{I}$ has $\mathbf{M}^8 = \mathbf{I}$. Find one with the right trace and determinant whose entries are not those of a rotation. How can you check that it is not a rotation or a reflection?

What's really going on

Similar matrices

Why is $\mathbf{M}$ in Stage 4 "a rotation in disguise"? Two matrices $\mathbf{M}$ and $\mathbf{R}$ are *similar* if $\mathbf{M} = \mathbf{S}\mathbf{R}\mathbf{S}^{-1}$ for some invertible $\mathbf{S}$: they represent the same transformation in different coordinates. Similar matrices have the same trace and determinant, and (for 2×2 matrices that are not multiples of $\mathbf{I}$) the converse holds too. So any real matrix with trace $\sqrt{2}$ and determinant 1 is similar to the rotation by $45°$, and in the skewed coordinates given by $\mathbf{S}$ it simply rotates (Figure 7.4.1 (in the full section online)). In general, a real 2×2 matrix with $\mathbf{M}^n = \mathbf{I}$ is always similar to a rotation or a reflection.

Why crystals have no five-fold symmetry

A crystal is a regular, repeating arrangement of atoms. A rotational symmetry of the crystal sends the lattice of atoms to itself, so in coordinates based on the lattice it is given by a matrix with *integer* entries. Its trace is therefore an integer. But a rotation by angle θ has trace $2\cos\theta$ in any coordinates, and $2\cos\theta$ is an integer only when it is $-2, -1, 0, 1$ or 2, that is, for rotations of order 2, 3, 4 and 6 (and 1). Five-fold symmetry would need $2\cos 72° = \frac{\sqrt{5}-1}{2}$, which is not an integer. This is the *crystallographic restriction theorem*. In 1982 Dan Shechtman observed an alloy with apparent five-fold symmetry and was at first ridiculed; it turned out to be a *quasicrystal*, ordered but not periodic, like the Penrose tilings, and so not covered by the theorem. He received the Nobel Prize in Chemistry in 2011.

Try this next

1. **Rotations of order three.** Try STEP II 2021, Question 7 (in OCR's download of the 2021 papers, www.ocr.org.uk/Images/696588-2021-past-papers.zip). It starts from a rotation matrix with an unusual property and ends with a fact about every real 2×2 matrix whose cube is the identity.

2. **Order six.** Find a condition on the trace and determinant for $\mathbf{M}^6 = \mathbf{I}$ but $\mathbf{M}^2 \neq \mathbf{I}$ and $\mathbf{M}^3 \neq \mathbf{I}$.

3. **The crystallographic restriction.** Show that if $\mathbf{M}$ has integer entries, $\det \mathbf{M} = 1$ and $\mathbf{M}^n = \mathbf{I}$ for some n, then $\operatorname{tr} \mathbf{M} \in \{-2, -1, 0, 1, 2\}$. Find an integer matrix of order 6.

4. **Cayley–Hamilton for powers.** Use $\mathbf{M}^2 = t\mathbf{M} - d\mathbf{I}$ to express $\mathbf{M}^3$ and $\mathbf{M}^4$ in the form $p\mathbf{M} + q\mathbf{I}$, and use this to rederive the condition for $\mathbf{M}^4 = -\mathbf{I}$.

Further reading

- **M. A. Armstrong, *Groups and Symmetry*** (Springer). Includes the crystallographic restriction and the classification of wallpaper patterns.
- **Marjorie Senechal, *Quasicrystals and Geometry*** (Cambridge University Press). The mathematics of quasicrystals and aperiodic order.
- **Gilbert Strang, *Introduction to Linear Algebra*** (Wellesley-Cambridge). Similar matrices and the Cayley–Hamilton theorem.

The full guided solution is in Section 7.4:
https://beyondstep.keithmcnulty.org/ch07/finite-order.html